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Department of Electronics and Communication Engineering

19EAC384 – Computer Networks Lab (S6 EAC)

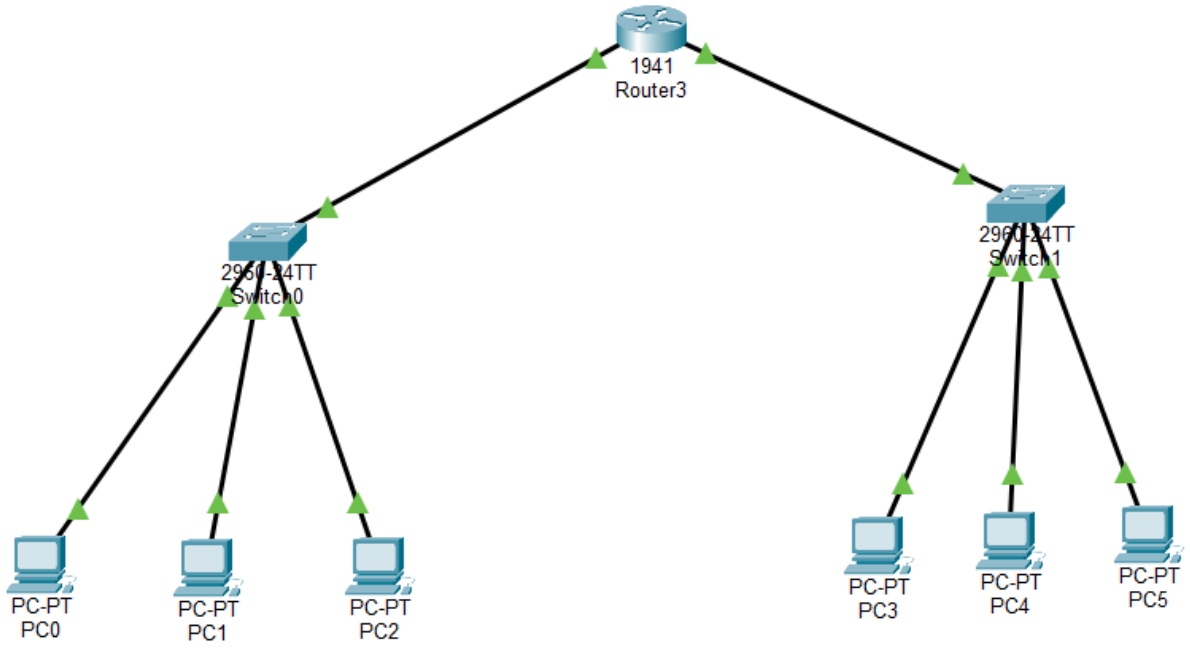
LABSHEET 4 - Assignment

BUILDING A WIDE AREA NETWORK (WAN) SET-UP

Aim:

- To build a WAN set up by configuring computers to form Local Area Networks (LAN) and establish communication between PCs in different LANs using Routers.
- To assign IP address dynamically to PCs using DHCP server.

Exercises-1: Build two simple LANs using 3 machines connected via star topology. Establish communication between two PCs belonging to different LANs using a single Router. In your document, highlight the steps adopted to set up the individual LANs and to configure the router and PCs to establish successful communication.



CLI Commands Listed:

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#! Assign IP address to interface
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.173.11.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! (Optional) Set another IP for second interface if needed
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.173.22.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#end
Router#
Router#! Save the configuration
Router#copy running-config startup-config
Destination filename [startup-config]?
%SYS-5-CONFIG_I: Configured from console by console
```

Ping Commands:

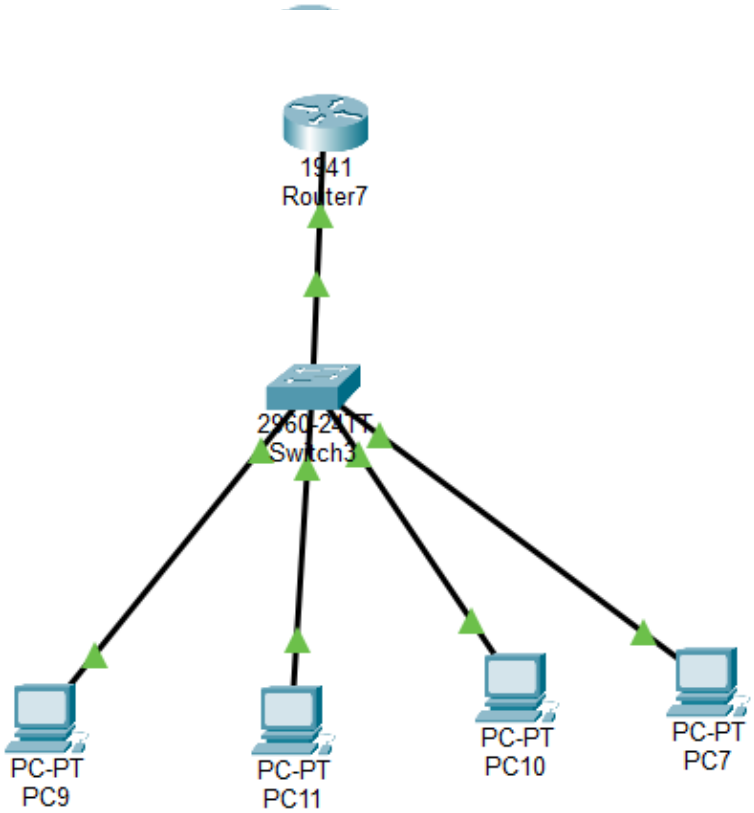
```
C:\>ping 192.173.22.2

Pinging 192.173.22.2 with 32 bytes of data:

Reply from 192.173.22.2: bytes=32 time<1ms TTL=128
Reply from 192.173.22.2: bytes=32 time<1ms TTL=128
Reply from 192.173.22.2: bytes=32 time=6ms TTL=128
Reply from 192.173.22.2: bytes=32 time=6ms TTL=128

Ping statistics for 192.173.22.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 3ms
```

```
C:\>ping 192.173.11.4

Pinging 192.173.11.4 with 32 bytes of data:

Reply from 192.173.11.4: bytes=32 time<1ms TTL=128
Reply from 192.173.11.4: bytes=32 time=4ms TTL=128
Reply from 192.173.11.4: bytes=32 time=4ms TTL=128
Reply from 192.173.11.4: bytes=32 time=3ms TTL=128

Ping statistics for 192.173.11.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 2ms
```

Inference:

Configuring two LANs connected via a router using CLI commands offered a valuable learning experience in understanding IP configuration and routing. Initially, I faced some difficulties in enabling communication between PCs on separate networks. However, after thoroughly checking the configuration and verifying that the IP addresses, subnet masks, and default gateways were correctly assigned, the issue was resolved, and inter-network communication was successfully achieved. This practical activity emphasized the critical role of accurate network settings in facilitating communication between devices across different LANs through a router.

Exercises-2: To configure a DHCP server in Cisco Packet Tracer that assigns IP addresses dynamically to 3 PCs connected to a switch. In your document give your observations.



CLI Commands Listed:

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.173.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.173.1.1 192.173.1.9
Router(config)#ip dhcp pool LAN_Pool
Router(dhcp-config)#network 192.173.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.173.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Ping Commands:

```
C:\>ping 192.173.1.11

Pinging 192.173.1.11 with 32 bytes of data:

Reply from 192.173.1.11: bytes=32 time<1ms TTL=128
Reply from 192.173.1.11: bytes=32 time<1ms TTL=128
Reply from 192.173.1.11: bytes=32 time<1ms TTL=128
Reply from 192.173.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.173.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.173.1.10

Pinging 192.173.1.10 with 32 bytes of data:

Reply from 192.173.1.10: bytes=32 time=11ms TTL=128
Reply from 192.173.1.10: bytes=32 time=5ms TTL=128
Reply from 192.173.1.10: bytes=32 time=3ms TTL=128
Reply from 192.173.1.10: bytes=32 time=4ms TTL=128

Ping statistics for 192.173.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 11ms, Average = 5ms
```

Inference:

Setting up DHCP using the command line on the router helped me understand how IP addresses are given out automatically in a network. At first, I had some trouble figuring out the right interface and how to create the DHCP pool correctly. But after trying different things and learning from my mistakes, I got it working. This experience made it easier to understand how IP addressing and router settings work, and it also made me feel more confident using the CLI in Cisco Packet Tracer.